



General Chemistry I

Recitation 6

Siyuan Chen
Nov. 3rd

Midterm feedback

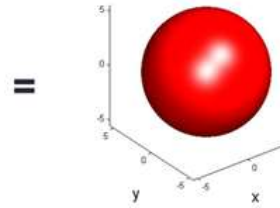
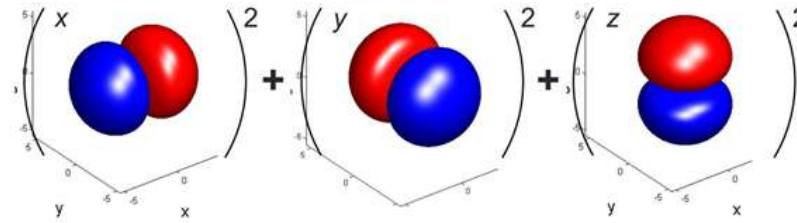
统计数据	
计数	63
最小值	3.20
最大值	10.00
范围	6.80
平均值	8.10
中位数	8.50
标准偏差	1.65
方差	2.73

Midterm feedback

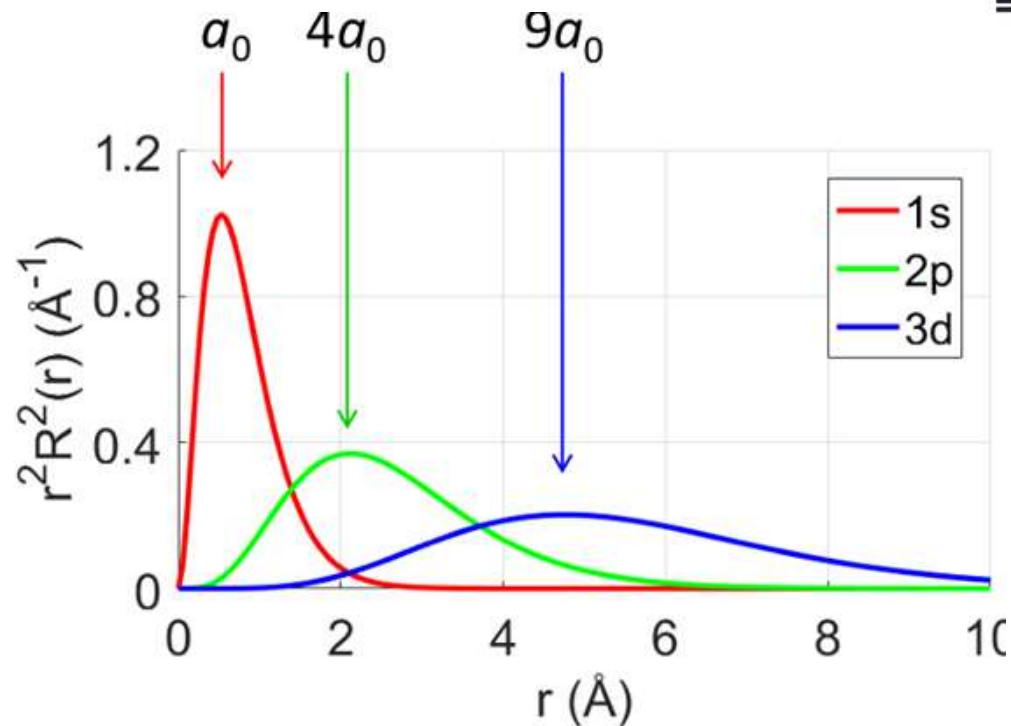
How to name a chemical properly?

Why Al_2Cl_6 stable?

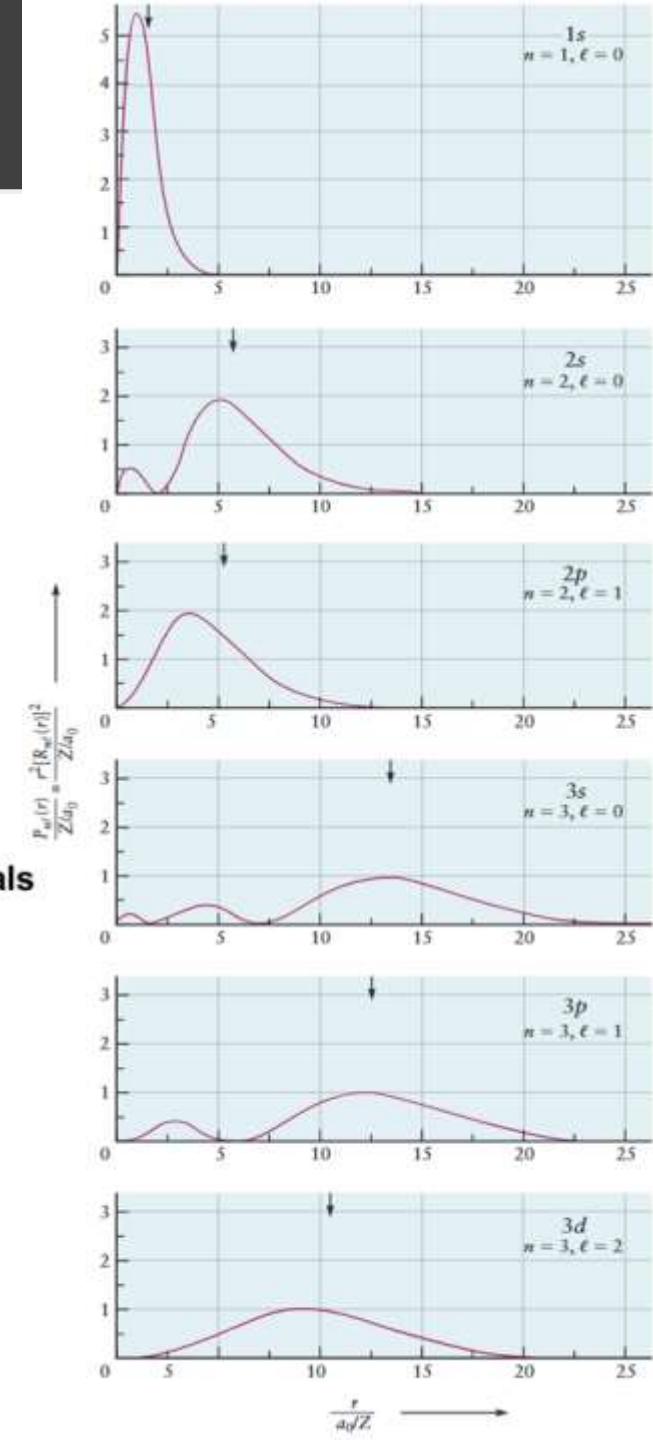
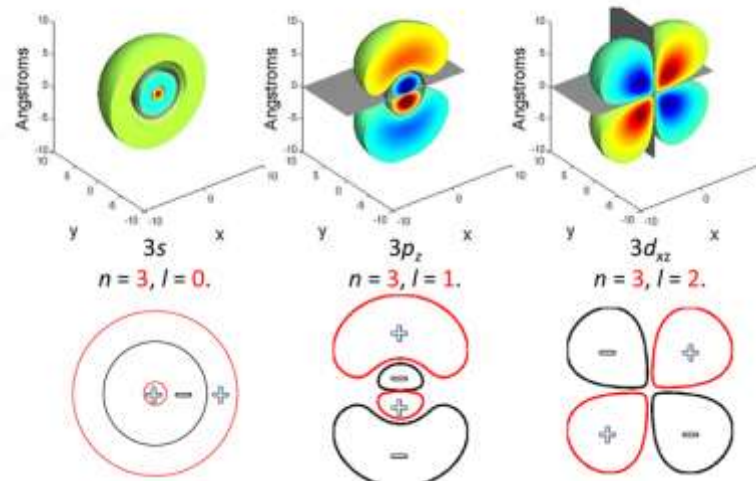
Lewis diagram..



The three p orbitals occupy
separate parts of the space.



The hydrogen atom: Comparison of 3s, 3p, 3d orbitals





Quiz5

可用性: 项目已不可用。上次可用日期为 2025-10-31 下午12:00。

1. Give the ground-state electron configurations of the following elements: (a) P (b) Tc (c) Ho

2. Identify the atom or ion corresponding to each of the following descriptions:

(a) an atom with ground-state electron configuration $[\text{Xe}] 4f^{14}5d^66s^2$

(b) an ion with charge -1 and ground-state electron configuration $[\text{He}] 2s^22p^6$

(c) an ion with charge +5 and ground-state electron configuration $[\text{Kr}] 4d^6$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																				
			6 7	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97		
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)		

Shielding Effect

Shielding effect sometimes referred to as atomic shielding or electron shielding describes the attraction between an electron and the nucleus in any atom with more than one electron.

Additionally, there is also a shielding effect that occurs between sublevels within the same principal energy level. An electron in the s-sublevel is capable of shielding electrons in the p-sublevel of the same principal energy level.

$$S(s) > S(p) > S(d) > S(f).$$

Slater's **empirical rules** for **estimating the shielding constant** for an electron in an ns or np orbital:

1. Write the electronic configuration for the atom with terms **in the order and grouped**: (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) etc.
2. **Electrons in any group to the right of a (ns, np) group do not shield the (ns, np) electrons** and therefore make no contribution to the shielding constant.
3. **Each of the other electrons in a (ns, np) group contributes 0.35** to the shielding constant for the valence electron.
4. **Each electron in the (n-1) shell contributes 0.85** to the shielding constant.
5. **Each electron in shells (n-2) or lower fully shield the outer electrons**, contributing 1.00 to the shielding constant.

$$V_n^{\text{eff}}(r) \approx -\frac{Z_{\text{eff}}(n)e^2}{r}$$

$$\epsilon_n \approx -\frac{[Z_{\text{eff}}(n)]^2}{n^2} \quad (\text{rydbergs})$$

$$\bar{r}_{n\ell} \approx \frac{n^2 a_0}{Z_{\text{eff}}(n)} \left\{ 1 + \frac{1}{2} \left[1 - \frac{\ell(\ell + 1)}{n^2} \right] \right\}$$

Shielding Effect

Slater's **empirical rules** for **estimating the shielding constant** for an electron in an ns or np orbital:

1. Write the electronic configuration for the atom with terms **in the order and grouped**: (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) etc.
2. **Electrons in any group to the right of a (ns, np) group do not shield the (ns, np) electrons** and therefore make no contribution to the shielding constant.
3. **Each of the other electrons in a (ns, np) group contributes 0.35** to the shielding constant for the valence electron.
4. **Each electron in the (n-1) shell contributes 0.85** to the shielding constant.
5. **Each electron in shells (n-2) or lower fully shield the outer electrons**, contributing 1.00 to the shielding constant.

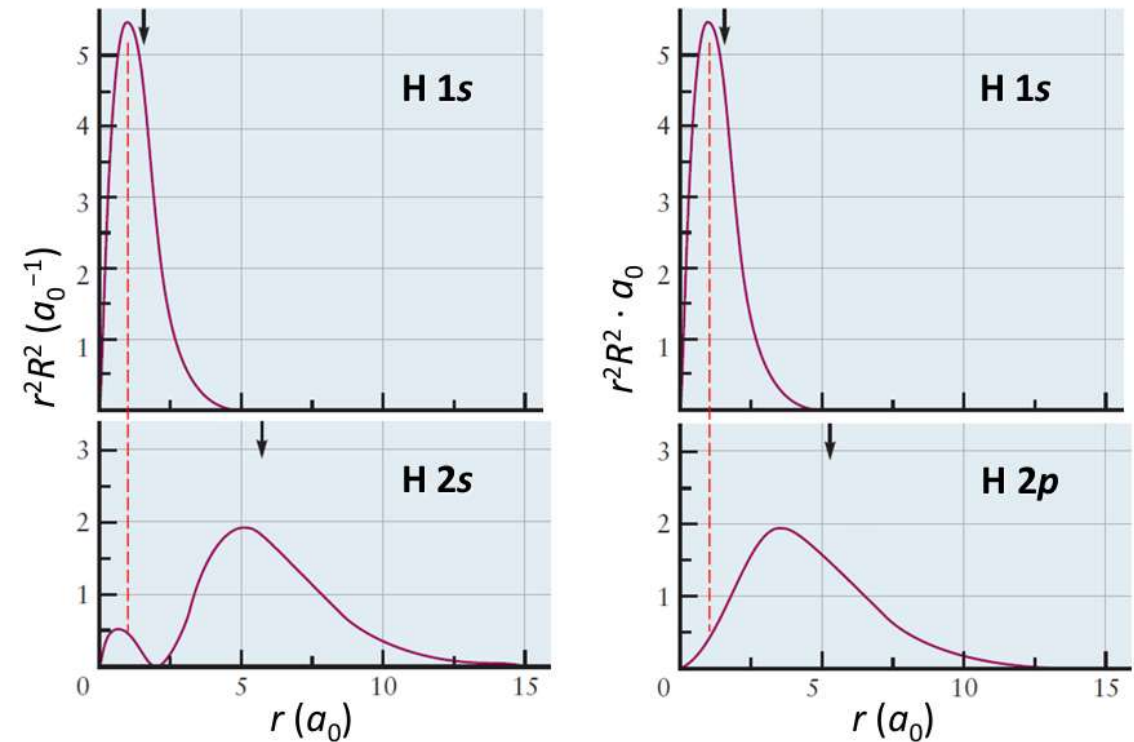
For an iron atom, the shielded (or effective) nuclear charge for each electron is deduced as:

$$\begin{array}{llllllll} 4s & : s = 0.35 \times 1 & + & 0.85 \times 14 & + & 1.00 \times 10 & = & 22.25 \Rightarrow Z_{\text{eff}}(4s) = 26.00 - 22.25 = 3.75 \\ 3d & : s = 0.35 \times 5 & & & + & 1.00 \times 18 & = & 19.75 \Rightarrow Z_{\text{eff}}(3d) = 26.00 - 19.75 = 6.25 \\ 3s, 3p & : s = 0.35 \times 7 & + & 0.85 \times 8 & + & 1.00 \times 2 & = & 11.25 \Rightarrow Z_{\text{eff}}(3s, 3p) = 26.00 - 11.25 = 14.75 \\ 2s, 2p & : s = 0.35 \times 7 & + & 0.85 \times 2 & & & = & 4.15 \Rightarrow Z_{\text{eff}}(2s, 2p) = 26.00 - 4.15 = 21.85 \\ 1s & : s = 0.30 \times 1 & & & & & = & 0.30 \Rightarrow Z_{\text{eff}}(1s) = 26.00 - 0.30 = 25.70 \end{array}$$

The Penetration Effect

Electrons with a higher probability of appearing near the atomic nucleus can better avoid repulsion from other electrons and experience stronger attraction from the nucleus, drawing them closer to it. This phenomenon, where electrons penetrate into the inner regions of the atom, is called the penetration effect. The penetration effect can lead to a lowering of energy levels.

$$E_{ns} < E_{np} < E_{nd} < E_{nf}$$



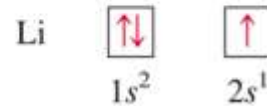
This phenomenon is known as energy level splitting. The reason lies in the distribution characteristics of the orbitals: among states with the same principal quantum number, the s-state has the highest number of peaks. Its distribution features a main peak farthest from the nucleus and smaller peaks closest to it. As the nuclear charge increases, the small peaks nearest to the nucleus play an increasingly significant role in energy. This is because the electron cloud represented by these small peaks effectively avoids shielding by other electrons, resulting in a higher effective nuclear charge acting on them. Additionally, this portion of the electron cloud is located closer to the nucleus. Both factors enhance the electron-nucleus interaction energy, significantly lowering the energy level of the orbital.

Aufbau principle and electron configuration

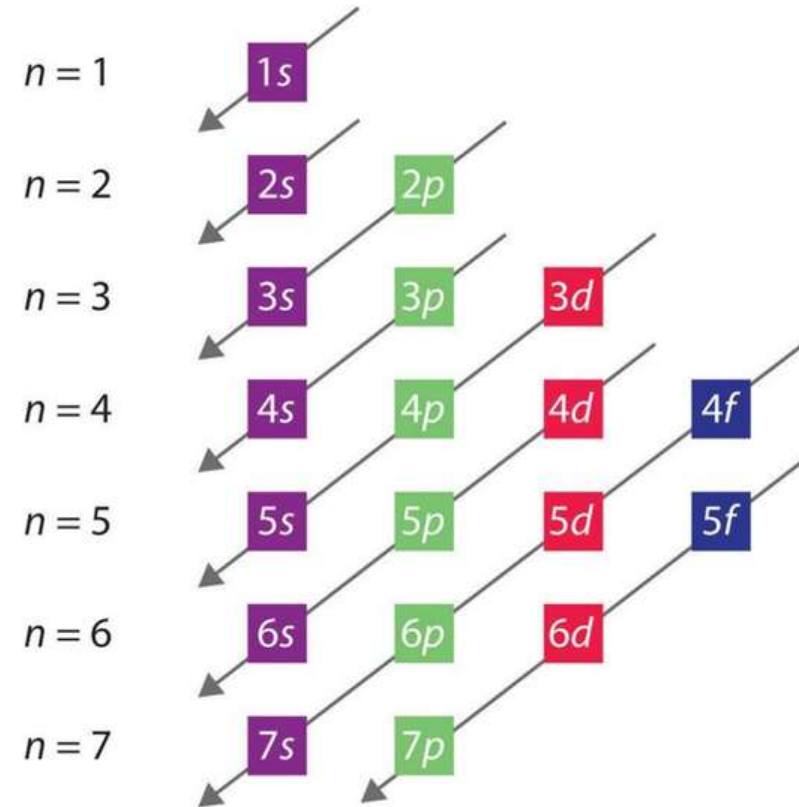
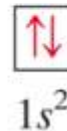
The **Pauli exclusion principle**: no two electrons in an atom can have the same set of four quantum numbers (n, ℓ, m, m_s).

Hund's rules: when electrons are added to Hartree orbitals of equal energy, a single electron enters each orbital before a second one enters any orbital. In addition, the lowest energy configuration is the one with parallel spins.

A substance is **paramagnetic** (顺磁性) if it is attracted into a magnetic field. Any substance that has one or more unpaired electrons in the atoms, molecules, or ions that compose it is paramagnetic because a net magnetic moment arises from each of the unpaired electrons.



A substance in which all the electrons are paired is weakly **diamagnetic** (反磁性). It is pushed out of a magnetic field, although the force it experiences is much smaller in magnitude than the force that pulls a typical paramagnetic substance into a magnetic field.



Aufbau principle and electron configuration

Optical spectroscopy confirms that the ground state **K** is $[\text{Ar}]3d^04s^1$ and that of **Ca** is $[\text{Ar}]3d^04s^2$, as predicted by the sequence of calculated orbital energies $\epsilon_{3d} > \epsilon_{4s}$

Why 4s?

For Sc and the elements beyond, advanced calculations predict that $\epsilon_{3d} < \epsilon_{4s}$

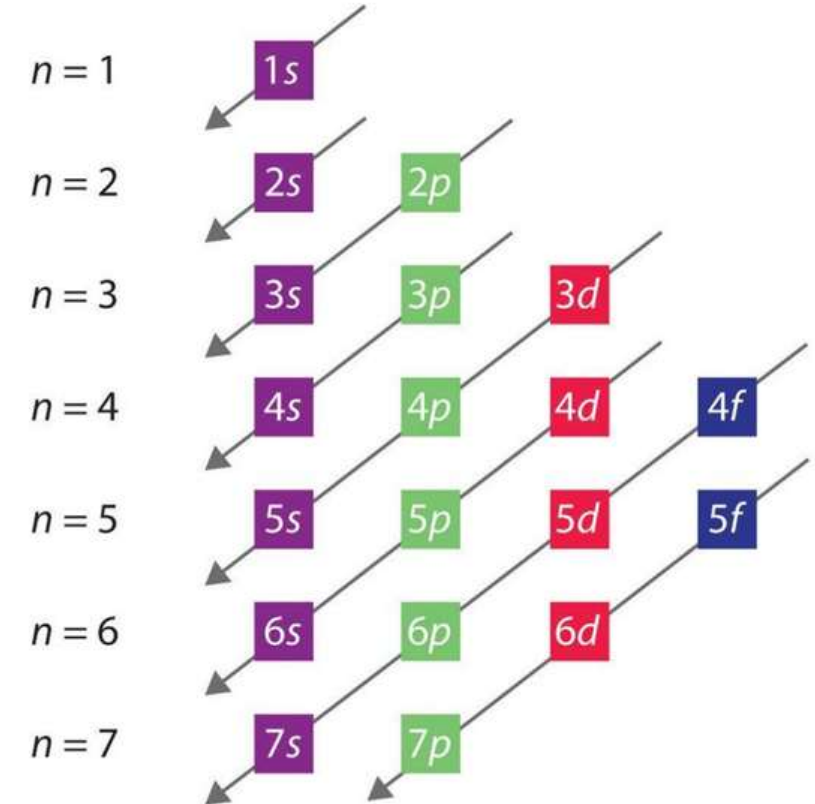
Filling the 3d orbitals first would give the configurations $[\text{Ar}]3d^34s^0$ for Sc, $[\text{Ar}]3d^44s^0$ for Ti, and so on to $[\text{Ar}]3d^{10}4s^0$ for Ni according to calculation. **However, experiments suggest $\epsilon_{3d} > \epsilon_{4s}$**



GUANGXIAN XU (徐光宪)

“ $n + 0.7 l$ ”

Try for 3d and 4s!

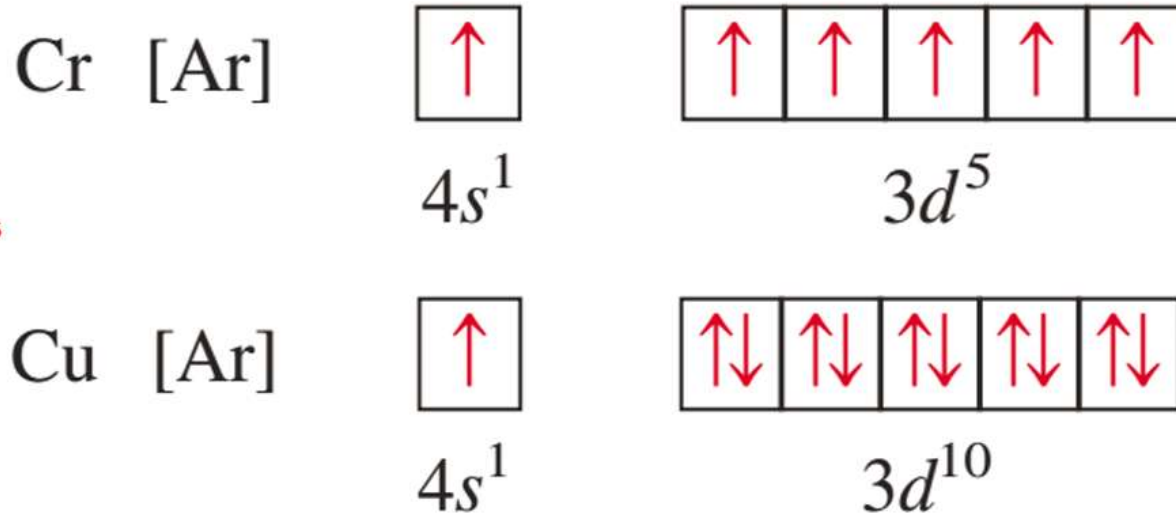


Aufbau principle and electron configuration

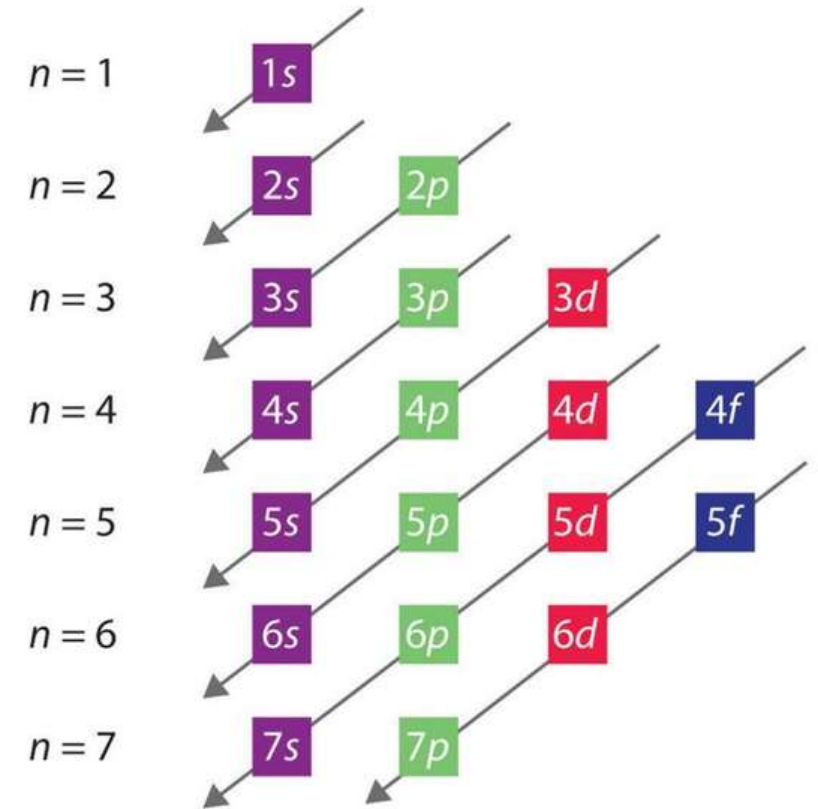
The configurations determined from calculations and experiment can be recalled as needed by assuming that the orbitals are filled in the sequence:

$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s \rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d$.

Why 4s?



Two exceptions



Lanthanide contraction（镧系收缩）

The lanthanide contraction is the greater-than-expected **decrease in atomic radii and ionic radii** of the elements in the lanthanide series, from left to right. It is caused by the **poor shielding effect of nuclear charge by the 4f electrons** along with the expected periodic trend of **increasing electronegativity and nuclear charge** on moving from left to right

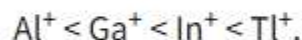
Element 元素	Atomic electron 原子电子 configuration 排布 (all begin with [Xe]) (所有都以[Xe]开头)	Ln ³⁺ electron Ln ³⁺ 电子 configuration 电子排布	Ln ³⁺ radius (pm) Ln ³⁺ 半径 (pm) (6-coordinate) (六配位)
La	5d ¹ 6s ²	4f ⁰	103
Ce	4f ¹ 5d ¹ 6s ²	4f ¹	102
Pr	4f ³ 6s ²	4f ²	99
Nd	4f ⁴ 6s ²	4f ³	98.3
Pm	4f ⁵ 6s ²	4f ⁴	97
Sm	4f ⁶ 6s ²	4f ⁵	95.8
Eu	4f ⁷ 6s ²	4f ⁶	94.7
Gd	4f ⁷ 5d ¹ 6s ²	4f ⁷	93.8
Tb	4f ⁹ 6s ²	4f ⁸	92.3
Dy	4f ¹⁰ 6s ²	4f ⁹	91.2
Ho	4f ¹¹ 6s ²	4f ¹⁰	90.1
Er	4f ¹² 6s ²	4f ¹¹	89
Tm	4f ¹³ 6s ²	4f ¹²	88
Yb	4f ¹⁴ 6s ²	4f ¹³	86.8
Lu	4f ¹⁴ 5d ¹ 6s ²	4f ¹⁴	86.1

“The effect results from poor shielding of nuclear charge (nuclear attractive force on electrons) by 4f electrons; the 6s electrons are drawn towards the nucleus, thus resulting in a smaller atomic radius.”

Inert-pair effect

The inert-pair effect is the tendency of the two electrons in the outermost atomic s-orbital to remain unshared in compounds of post-transition metals. The term inert-pair effect is often used in relation to the increasing stability of oxidation states that are two less than the group valency for the heavier elements

Consider as an example thallium (Tl) in [group 13](#). The +1 oxidation state of Tl is the most stable, while Tl^{3+} compounds are comparatively rare. The stability of the +1 oxidation state increases in the following sequence:^[2]



The same trend in stability is noted in groups [14](#), [15](#) and [16](#). The heaviest members of each group, i.e. [lead](#), [bismuth](#) and [polonium](#) are comparatively stable in oxidation states +2, +3, and +4 respectively.

Ionization energies for group 13 elements

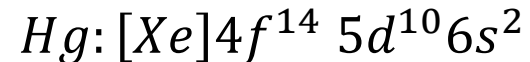
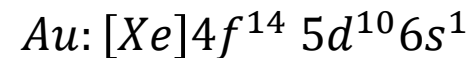
	kJ/mol				
IE	Boron	Aluminium	Gallium	Indium	Thallium
1st	800	577	578	558	589
2nd	2427	1816	1979	1820	1971
3rd	3659	2744	2963	2704	2878
2nd + 3rd	6086	4560	4942	4524	4849

“The effect results from poor shielding of nuclear charge (nuclear attractive force on electrons) by 4f electrons; the 6s electrons are drawn towards the nucleus.”

Some interesting phenomena

Differences in Properties between Gold and Mercury

Gold and mercury differ by only one electron. Their electron configurations are as follows



Cl-Hg-Hg-Cl

Due to relativistic effects, the 6s orbital contracts and its energy level decreases. The 6s and 5d orbitals together form the outermost valence orbitals. In this state, gold has an electron configuration analogous to that of the halogens (lacking one electron from a closed shell). Some of its chemical properties are similar to those of halogens. For example, gold can form covalent single bonds to produce Au_2 molecules, which exist in the gas phase (like halogen X_2 molecules); gold can form ionic compounds such as $RbAu$ and $CsAu$, in which gold is present as the monovalent negative ion Au^- , similar to halogens (X) gaining one electron to form a stable closed-shell structure.